

```
// Structural description of design example (See Figs. 8.9(a), 8.12)
module Design_Example_STR
  ( output [3: 0] A,                      // V 2001 port syntax
    output E, F,
    input Start, clock, reset_b
  );
  Controller_STR M0 (clr_A_F, set_E, clr_E, set_F, incr_A, Start, A[2], A[3], clock,
reset_b );
  Datapath_STR M1 (A, E, F, clr_A_F, set_E, clr_E, set_F, incr_A, clock);
endmodule
```

```
// Control Unit
module Controller_STR
  ( output clr_A_F, set_E, clr_E, set_F, incr_A,
    input Start, A2, A3, clock, reset_b
  );
  wire G0, G1;
  parameter S_idle = 2'b00, S_1 = 2'b01, S_2 = 2'b11;
  wire w1, w2, w3;
  not (G0_b, G0);
  not (G1_b, G1);
  buf (incr_A, w2);
  buf (set_F, G1);
  not (A2_b, A2);
  or (D_G0, w1, w2);
  and (w1, Start, G0_b);
  and (clr_A_F, G0_b, Start);
  and (w2, G0, G1_b);
  and (set_E, w2, A2);
  and (clr_E, w2, A2_b);
  and (D_G1, w3, w2);
  and (w3, A2, A3);
  D_flip_flop_AR M0 (G0, D_G0, clock, reset_b);
  D_flip_flop_AR M1 (G1, D_G1, clock, reset_b);
endmodule
```

```
// datapath unit
module Datapath_STR
  ( output [3: 0] A,
    output E, F,
    input clr_A_F, set_E, clr_E, set_F, incr_A, clock
  );
  JK_flip_flop_2 M0 (E, E_b, set_E, clr_E, clock);
  JK_flip_flop_2 M1 (F, F_b, set_F, clr_A_F, clock);
  Counter_4 M2 (A, incr_A, clr_A_F, clock);
endmodule
```

```
// Counter with synchronous clear
module Counter_4 (output reg [3: 0] A, input incr, clear, clock);
```

```

always @ (posedge clock)
    if (clear) A <= 0; else if (incr) A <= A + 1;
endmodule

module D_flip_flop_AR (Q, D, CLK, RST); // Asynchronous reset
    output    Q;
    input     D, CLK, RST;
    reg       Q;
    always @ (posedge CLK, negedge RST)
        if (RST == 0) Q <= 1'b0;
        else Q <= D;
endmodule

// Description of JK flip-flop
module JK_flip_flop_2 (Q, Q_not, J, K, CLK);
    output    Q, Q_not;
    input     J, K, CLK;
    reg       Q;
    assign    Q_not = ~Q;
    always @ (posedge CLK)
        case ({J, K})
            2'b00: Q <= Q;           // No change
            2'b01: Q <= 1'b0;        // Clear
            2'b10: Q <= 1'b1;        // Set
            2'b11: Q <= ~Q;          // Toggle
        endcase
endmodule

// Tests Bench
module t_Design_Example_STR;
    reg Start, clock, reset_b;
    wire [3: 0] A;
    wire     E, F;
    // Instantiate design example
    Design_Example_STR M0 (A, E, F, Start, clock, reset_b);
    // Describe stimulus waveforms
    initial #500 $finish;           // Stopwatch
    initial
        begin
            reset_b = 0;
            Start = 0;
            clock = 0;
            #5 reset_b = 1; Start = 1;
            repeat (32)
                begin
                    #5 clock = ~ clock;    // Clock generator
                end
            end
        initial
            $monitor ("A = %b E = %b F = %b time = %0d", A, E, F, $time);

```

endmodule
